

AI LAB PROJECT REPORT

Title:

ReceiptIQ – Privacy-First Personal Finance Assistant

Submitted By:

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Role: Machine Learning Engineer

1. Introduction

This project is developed as a group to automate expense tracking using AI.

ReceiptIQ extracts data from receipts and provides insights while ensuring privacy through local processing.

2. System Overview

The system consists of multiple modules:

- OCR for text extraction
 - Data preprocessing
 - Machine learning classification
 - Backend storage
 - Analytics dashboard
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3. Machine Learning Module

The ML module is responsible for categorizing expenses.

Approach:

We used **TF-IDF** to convert text into numerical features.

Models:

- SVM
- Random Forest

Focus:

- Accuracy
 - Speed
 - Real-world performance
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4. Challenges

- Noisy and short text
- Irregular formats
- Abbreviations

Solution:

- Text cleaning
 - Feature optimization
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5. My Contribution

- Built classification pipeline
 - Implemented TF-IDF
 - Trained and evaluated models
 - Optimized performance
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6. Conclusion

The ML module enables automatic expense categorization, reducing manual effort and improving usability.